HUNDRED DAYS OF CODE : DAY - 7

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1.Problem Statement - 1

Write a C program to find escape velocity when mass and radius of a planet are given.

1.1 Inputs, Outputs & Formula Used

The Formula used for this problem is : Ve = sqrt (2GM/R)

G = gravitational constant
Mass of the planet (M) and its radius (R)

The overview of the program is given as :

Test Case	Input	Expected Output	Actual Output	Result
1.Earth	M: 5.972e24, R: 6371000	11185.80 m/s	11185.80 m/s	PASS
2.Moon	M: 7.342e22, R: 1737400	2374.88 m/s	2374.88 m/s	PASS

1.2 The algorithm

1. Start.
2. Declare variables for mass, radius, and escape_velocity.
3. Define the constant G = 6.674 \times 10^{-11}.
4. Prompt the user to input the planet's mass and radius.
5. Calculate escape_velocity = sqrt((2 * G * mass) / radius).
6. Print the escape_velocity.
7. Stop.

1.3 The C Program

```
#include <stdio.h>
#include <math.h>

int main()
{
   double mass, radius, escape_velocity;
   const double G = 6.674e-11;

   printf("Enter mass of the planet (kg): ");
   scanf("%lf", &mass);

   printf("Enter radius of the planet (m): ");
   scanf("%lf", &radius);

   escape_velocity = sqrt((2 * G * mass) / radius);

   printf("Escape Velocity: %.2lf m/s\n",
escape_velocity);
   return 0;
}
```

1.4 Output

```
┌──(kali㉿kali)-[~/codes]
└─$ gcc que1.c -o que1 -lm

┌──(kali㉿kali)-[~/codes]
└─$ ./que1
Enter mass of the planet (kg): 5.972e24
Enter radius of the planet (m): 6371000
Escape Velocity: 11185.73 m/s

┌──(kali㉿kali)-[~/codes]
└─$ ./que1
Enter mass of the planet (kg): 7.342e22
Enter radius of the planet (m): 1737400
Escape Velocity: 2375.01 m/s

┌──(kali㉿kali)-[~/codes]
└─$ 
```

2.Problem Statement - 11

Write a C program to find height energy per unit volume of a fluid when density, gravity, and height are given.

2.1 Inputs, Outputs & Formula Used

The Formulas used for this problem is : 1. PE = (rho)*g*h

where, rho = density
g = acceleration due to gravity
h = height from the reference point

The overview of the program is given as :

Test Case	Input	Expected Output	Actual Output	Result
1.	rho:1.8 g:9.8 h:10	176.4	176.4	PASS
2.	rho:2.0 g:9.8 h:25	490	490	PASS
3.	rho:2.4 g:9.6 h:22	506.8	506.8	PASS

2.2 The algorithm

1. Start.
2. Declare variables for density, gravity, height, and energy.
3. Read inputs for density, gravity, and height.
4. Calculate energy = density * gravity * height.
5. Print the calculated energy.
6. Stop.

2.3 The C Program

```
#include<stdio.h>

int main()
{
 float rho, g, h, PE;

 printf("Enter the density of fluid: \n");
 scanf("%f",&rho);
 printf("Enter the acceleration due to gravity: \n");
 scanf("%f",&g);
 printf("Enter the height from the reference point: \n");
 scanf("%f",&h);

 PE = rho*g*h;

 printf("The energy per unit volume is: %f\n",PE);
 return 0;
}
```

2.4 Output

```
┌──(kali㉿kali)-[~/codes/day_7]
└─$ gcc que2.c -o que2

┌──(kali㉿kali)-[~/codes/day_7]
└─$ ./que2
Enter the density of fluid:
1.8
Enter the acceleration due to gravity:
9.8
Enter the height from the reference point:
10
The energy per unit volume is: 176.399994

┌──(kali㉿kali)-[~/codes/day_7]
└─$ ./que2
Enter the density of fluid:
2.0
Enter the acceleration due to gravity:
9.8
Enter the height from the reference point:
25
The energy per unit volume is: 490.000000

┌──(kali㉿kali)-[~/codes/day_7]
└─$ ./que2
Enter the density of fluid:
2.4
Enter the acceleration due to gravity:
9.6
Enter the height from the reference point:
22
The energy per unit volume is: 506.880005
```

3.Problem Statement - III

Write a C program to find photon energy when frequency is given.

3.1 Inputs, Outputs & Formula Used

The Formulas used for this problem is : 1. E = hf

where, E = energy of photon
h = plank's constant
f = frequency

The overview of the program is given as :

Test Case	Input	Expected Output	Actual Output	Result
1.	f: 7.5e14	4.770000e-19	4.770000e-19	PASS
2.	f: 1.5e18	9.540000e-16	9.540000e-16	PASS
3.	f:2.2e-25	1.399200e-58	1.399200e-58	PASS

2.2 The algorithm

1. Start.
2. Declare a variable for plank, frequency and energy.
3. Read the frequency input.
4. Define the plank's constant = 6.36*10^{-34}
5. Calculate energy = Planck_h * frequency.
6. Print the energy using scientific notation formatting.
7. Stop.

2.3 The C Program

```
#include<stdio.h>

int main()
{
 double E, h, f;

 printf("Enter the frequency of the EM wave: \n");
 scanf("%lf",&f);

 h = 6.36e-34;
 E = h*f;

 printf("The photon energy is: %e\n",E);
 return 0;
}
```

3.4 Output

```
┌──(kali㉿kali)-[~/codes/day_7]
└─$ gcc que3.c -o que3

┌──(kali㉿kali)-[~/codes/day_7]
└─$ ./que3
Enter the frequency of the EM wave:
7.5e14
The photon energy is: 4.770000e-19

┌──(kali㉿kali)-[~/codes/day_7]
└─$ ./que3
Enter the frequency of the EM wave:
1.5e18
The photon energy is: 9.540000e-16

┌──(kali㉿kali)-[~/codes/day_7]
└─$ ./que3
Enter the frequency of the EM wave:
2.2e-25
The photon energy is: 1.399200e-58
```